IMAGE PROCESSING & COMPUTER VISION (IPCV) PROJECT REPORT

Focus-Analyser

Attention Span & Productivity Monitoring System

A computer vision framework for real-time attention, drowsiness and distraction monitoring

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ABSTRACT

Sustained attention is important in digital learning and remote work, yet common activity trackers mainly record keyboard, mouse or application use. These signals cannot reliably show whether a person is visually engaged, looking away, becoming drowsy or interacting with a distracting object. Focus-Analyser addresses this gap through a webcam-based computer vision system that estimates attention-related behaviour without requiring specialist wearable sensors.

The system processes video frames using OpenCV and facial landmarks from Dlib. It combines eye openness and blink dynamics (Eye Aspect Ratio, EAR), gaze direction, 3D head orientation using the Perspective-n-Point (solvePnP) method, eye-closure fatigue (PERCLOS), yawning (Mouth Aspect Ratio, MAR), upper-body posture and object detection using YOLOv8. These signals are fused into an attention score from 0 to 100, then summarized into behavioural categories. Focus durations, streaks and session-level statistics are stored in SQLite and presented through visual summaries.

The supplied project report records evaluation across 20 sessions and 1.81 hours of runtime. It reports an average processing throughput of 29.4 frames per second, a mean attention score of 49.9%, and a longest continuous focus streak of 412 seconds. These results indicate the feasibility of near-real-time monitoring on standard CPU-based hardware, while further testing is needed to establish general accuracy across users and environments.

Keywords: Computer Vision, Attention Estimation, EAR, solvePnP, PERCLOS, MAR, YOLOv8, SQLite, Real-Time Monitoring.

1. INTRODUCTION

1.1 Background and Motivation

Online education, remote work and long periods of screen-based activity make it difficult for people to judge how consistently they are concentrating. Frequent context switching and small interruptions can break a work session, while a person may remain physically present in front of a computer without actively attending to the task. Conventional activity trackers observe computer interaction but do not directly interpret facial or posture cues.

Specialist equipment such as EEG headsets and infrared eye trackers can provide additional physiological information, but they may be expensive, intrusive or inconvenient for everyday use. Focus-Analyser therefore uses a standard webcam to estimate observable indicators of attention and fatigue. Its output is intended as a productivity aid, not as a clinical diagnosis or a definitive measurement of a person's internal mental state.

1.2 Problem Statement and Objectives

The project aims to move beyond simple face-present/face-absent detection. Its main challenges are changes in lighting and facial appearance, natural differences in resting posture, the need to process video quickly, and the need to explain why an attention score has fallen. The system's objectives are to extract several visual cues, combine them into a comprehensible score, track focus over time, and retain session summaries for later review.

2. METHODOLOGY

2.1 System Workflow

The system follows a modular pipeline. Frames are captured from a webcam, resized to 640 × 480 and converted to grayscale for facial processing. Dlib's HOG-based detector locates the face, and a 68-point facial landmark model estimates key positions around the eyes, nose, mouth and jaw. The landmarks are passed to specialized analyzers, while RGB frames are also sampled by the YOLOv8 object detector. Their outputs are combined, smoothed over time and saved with session summaries.

Stage	Purpose
1. Frame capture	Read webcam frames with OpenCV; resize and prepare the image.
2. Face landmarks	Detect the face and locate 68 facial reference points.
3. Feature analysis	Estimate eye closure, gaze, head pose, yawning and posture.
4. Object detection	Check sampled frames for objects such as phones or books.
5. Score and tracking	Fuse feature values, classify state and track focus streaks.
6. Storage and reports	Save session metrics in SQLite for charts and later review.

2.2 Facial and Eye Measurements

The 68 facial landmarks provide image coordinates for the geometric calculations. Eye Aspect Ratio (EAR) compares vertical eyelid separation with eye width. A low EAR suggests that the eye is closed; short closures are treated as blinks, while prolonged closure can trigger a drowsiness or microsleep alert. The project uses an EAR threshold of 0.22 and treats closure lasting more than 15 frames as prolonged at the target frame rate.

Gaze estimation examines the relative position and intensity pattern of the pupil and surrounding sclera in the cropped eye region. The resulting horizontal gaze ratio is used to identify possible peripheral looking. Because ordinary webcams and visible-light images are affected by lighting, glasses and pupil visibility, gaze should be interpreted as an estimate rather than a perfect eye-tracking measurement.

2. METHODOLOGY (CONTINUED)

2.3 3D Head Pose Estimation

Head pose estimates the orientation of the face in three dimensions. The implementation selects six facial landmarks and matches their 2D image positions to corresponding points in a simplified 3D head model. OpenCV's solvePnP algorithm uses these correspondences and camera parameters to estimate rotation and translation. The rotation is converted into Euler angles: yaw describes turning left or right, pitch describes looking up or down, and roll describes tilting sideways.

At the beginning of a session, the system records a short resting baseline so that a user's normal camera angle is not automatically treated as distraction. The report specifies alerts when yaw differs from baseline by more than 28° or pitch by more than 25°. This calibration is particularly important for laptop users whose normal viewing angle may be downward.

2.4 Fatigue, Yawning and Object Detection

PERCLOS estimates the proportion of a recent time window during which the eyes are substantially closed. The implementation uses a 100-frame window (about 3.3 seconds at 30 FPS); a value above 20% is treated as a drowsiness indicator. Mouth Aspect Ratio (MAR) compares mouth opening with mouth width. A MAR above 0.60 sustained for more than 20 frames is used to flag a possible yawn. Upper-body posture displacement provides an additional ergonomic cue.

For environmental distraction, the project uses the lightweight YOLOv8n object detector. It samples frames asynchronously and checks for items such as cell phones and books at a confidence threshold of 0.35. An identified object can be highlighted in the interface and reduce the composite attention score. Detection is limited to the classes supported by the configured model and does not prove that an object is actually being used.

2.5 Composite Attention Score

Each analyzer contributes a normalized quality value between 0 and 1. The weighted values are combined into a score between 0 and 100, with an additional penalty when a physical distractor is detected. The weights stated in the project are: gaze 0.25, head pose 0.25, PERCLOS 0.20, posture 0.15 and yawning 0.15; the distractor penalty is 0.40. An exponential moving average with $\alpha = 0.15$ reduces rapid score fluctuations.

Output category	Score range stated in report
Focused	75–100%
Moderate	50–74%
Distracted	25–49%
Drowsy	Below 25%

The system also tracks continuous focus spans while the score is at least 65%. If the score remains below that threshold for more than three seconds, the current streak ends. Session metrics are grouped into five-second intervals to create a focus heatmap. SQLite stores session duration, average score, time spent in each state, longest streak, blink and yawn counts, and the primary recorded distractor.

Implementation note: these values are the thresholds and weights documented in the supplied project report. Their suitability depends on camera setup and calibration; they should not be interpreted as universally validated clinical thresholds.

3. RESULTS AND DISCUSSION

3.1 Evaluation Setup

According to the supplied report, evaluation covered 20 user sessions with a combined runtime of 1.81 hours (6,516 seconds). Tests involved 11th-generation Intel Core i5/i7 CPUs without external GPU acceleration, standard integrated 720p webcams, and illumination ranging from approximately 150 to 650 lux. Activities included reading academic material, coding, typing and watching instructional videos.

3.2 Processing Performance

Pipeline component	Reported time	Share
Frame acquisition and mirroring	3.1 ms	9.1%
Face detection	12.4 ms	36.5%
Facial landmark alignment	7.8 ms	22.9%
Eye and gaze estimation	1.9 ms	5.6%
3D head pose	1.6 ms	4.7%
YOLOv8n distractor detection	6.2 ms	18.2%
Scoring and interface rendering	1.0 ms	3.0%
Total end-to-end	34.0 ms / 29.4 FPS	100%

The reported end-to-end latency is 34.0 milliseconds per frame, equivalent to about 29.4 frames per second. Face detection is the largest listed processing cost, followed by facial landmark alignment and object detection. The results suggest that the selected combination of algorithms can operate close to a standard 30 FPS webcam rate on CPU-based hardware. Actual speed may vary with processor, image conditions, model settings and background load.

3.3 Session-Level Findings

Metric	Reported result	Interpretation
Evaluated sessions	20	Multiple sessions were recorded.
Total monitored time	1.81 hours	Provides a limited real-world runtime sample.
Mean attention score	49.9% ± 18.4%	Sessions included a mixture of focus and distraction.
Maximum session score	80.0%	Recorded during sustained reading.
Minimum session score	9.0%	Recorded during eye closure and phone distraction.
Average focus streak	54.2 ± 28.6 seconds	Typical uninterrupted focus interval in this sample.
Longest focus streak	412 seconds	Demonstrates tracking of a longer concentration period.
Natural blink frequency	18.4 ± 4.2 blinks/min	Within the report's cited typical resting range.

These figures describe the sessions included in the project report; they are not a controlled clinical validation or a guarantee of accuracy for every user. The average score is influenced by the types of tasks and intentional distraction tests included in the sessions. A larger and more diverse evaluation would be needed to estimate false-positive and false-negative rates.

3. RESULTS AND DISCUSSION (CONTINUED)

3.4 Distractor Attribution

Primary session label	Sessions	Share
Eyes closed / drowsiness	10	50%
Head turned away	8	40%
Yawning	1	5%
Uninterrupted high focus	1	5%

The report identifies eye closure/drowsiness and head turning as the most frequent primary labels in the evaluated sessions. It also notes that natural downward laptop viewing initially caused false head-turn alerts. Baseline angle subtraction and a wider pitch tolerance were introduced to reduce these false positives. This illustrates why personal calibration is important, although the report's claim of complete elimination should be confirmed with a larger independent test set.

3.5 Limitations and Implications

Focus-Analyser offers a practical way to summarize visible attention-related cues using widely available hardware. Its main strengths are the combination of several signals, near-real-time operation, interpretable alert categories and persistent session summaries. However, looking away does not always mean a person is distracted, and looking at the screen does not guarantee mental engagement. Lighting, spectacles, camera placement, facial occlusion, individual behaviour and the detector's supported object classes can affect results. The attention score is therefore best treated as a self-monitoring indicator rather than an objective measurement of cognition.

4. CONCLUSION

Focus-Analyser combines facial landmark geometry, eye and gaze analysis, 3D head pose estimation, fatigue indicators and object detection into a webcam-based attention monitoring pipeline. It converts these observations into a composite score, tracks focus streaks and stores session-level information in SQLite for later review. The supplied evaluation reports 29.4 FPS throughput across 20 sessions and 1.81 hours of operation, demonstrating the technical feasibility of near-real-time processing on standard CPU hardware.

The project provides a useful foundation for self-regulated learning, productivity reflection and ergonomic awareness. Future work can strengthen it by testing more participants and lighting conditions, measuring precision and recall against labelled ground truth, improving user-specific gaze calibration, evaluating temporal models for behaviour sequences, supporting multiple people, and exploring mobile or tablet deployment. Privacy-conscious handling of video and clear communication that the score is an estimate should remain central to further development.

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